

7 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE

KEY FINDINGS

- The amplicon sequencing segment accounted for the largest share of 41.0% of the global microbiome sequencing services market in 2022. This segment is projected to reach USD 234.18 million by 2028 from USD 118.98 million in 2023, at a CAGR of 14.5%.
- The shotgun sequencing segment is expected to grow at the highest CAGR of 15.2% from 2023 to 2028. This segment is expected to reach a value of USD 157.10 million by 2028.
- Whole-genome sequencing is the third-largest segment in terms of revenue, with a market share of approximately 21.0% in 2022.

7.1 INTRODUCTION

Based on type, the global microbiome sequencing services market is segmented into amplicon sequencing, whole-genome sequencing, shotgun sequencing, and other types (including de novo sequencing, metagenomic sequencing, and metatranscriptomic sequencing). The amplicon sequencing segment accounted for the largest market share of 41.0% in 2022. The large share of this segment can be attributed to the high adoption of 16S rRNA microbiome sequencing and targeted sequencing. Key market players/service providers have built an innovative portfolio around these services, including sample preparation and data analysis solutions.

TABLE 29 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	91.80	104.54	118.98	135.54	154.55	176.84	203.06	234.18	14.5%
Whole-genome Sequencing	47.29	53.48	60.45	68.39	77.44	87.98	100.31	114.85	13.7%
Shotgun Sequencing	59.09	67.70	77.52	88.83	101.89	117.27	135.44	157.10	15.2%
Other Types	21.85	24.46	27.36	30.62	34.29	38.52	43.41	49.11	12.4%
Total	220.03	250.17	284.31	323.39	368.16	420.61	482.21	555.24	14.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.2 AMPLICON SEQUENCING

7.2.1 HIGH ACCURACY OF 16S AMPLICON SEQUENCING TO DRIVE ADOPTION

This segment is inclusive of 16S rRNA sequencing, ITS sequencing, and targeted sequencing services. Amplicon sequencing is a highly targeted approach that allows researchers to analyze genetic variation in specific regions of a genome. The ultra-deep sequencing of PCR products (amplicons) allows efficient variant identification and characterization. 16S rRNA sequencing is the most common technique utilized for metagenomics studies. In this technique, the 16S rRNA genetic markers present in a bacterial genome are sequenced to study bacterial phylogeny and taxonomy from complex microbiomes or environments that are difficult or impossible to analyze. The presence of 16S rRNA genetic markers as a multigene family or operons in almost all bacteria and the large size of the 16S rRNA gene (~1,500 bp) makes it the most preferred metagenomic sequencing technique among end users.

The demand for targeted sequencing is anticipated to increase in metagenomic research activities. The targeted amplicon sequencing method involves sequencing a phylogenetically informative marker to identify organisms in community samples. The marker must be present across all samples with expected organisms to enhance taxonomic profiling through the amplification of the primer. A number of different markers are commonly used and vary by the taxonomy of interest, but the most commonly used targeted amplicon approaches include the 16S rRNA gene for bacterial samples, the 18S for eukaryotes, and ITS for fungal samples.

TABLE 30 AMPLICON SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	52.63	59.60	67.44	76.40	86.62	98.54	112.50	128.99	13.8%
Europe	28.39	32.47	37.11	42.44	48.59	55.82	64.35	74.51	15.0%
Asia Pacific	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%
Latin America	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%
Middle East and Africa	1.22	1.35	1.48	1.63	1.79	1.97	2.17	2.40	10.1%
Total	91.80	104.54	118.98	135.54	154.55	176.84	203.06	234.18	14.5%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 31 NORTH AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	47.70	54.05	61.22	69.40	78.74	89.64	102.42	117.52	13.9%
Canada	4.93	5.54	6.22	7.00	7.88	8.90	10.08	11.48	13.0%
Total	52.63	59.60	67.44	76.40	86.62	98.54	112.50	128.99	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 32 EUROPE: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	6.16	7.06	8.08	9.26	10.62	12.22	14.11	16.37	15.2%
UK	5.50	6.34	7.29	8.40	9.68	11.20	13.00	15.15	15.7%
France	4.34	4.97	5.68	6.50	7.45	8.56	9.87	11.43	15.0%
Italy	1.22	1.40	1.59	1.82	2.08	2.38	2.74	3.16	14.7%
Spain	0.94	1.07	1.22	1.39	1.58	1.81	2.08	2.40	14.5%
Rest of Europe	10.22	11.64	13.24	15.08	17.18	19.65	22.55	25.99	14.4%
Total	28.39	32.47	37.11	42.44	48.59	55.82	64.35	74.51	15.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 33 ASIA PACIFIC: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.30	3.88	4.55	5.34	6.27	7.39	8.72	10.34	17.8%
Japan	1.53	1.81	2.14	2.53	2.98	3.53	4.20	5.00	18.5%
India	0.19	0.23	0.27	0.31	0.37	0.43	0.51	0.60	17.7%
Rest of Asia Pacific	2.76	3.24	3.79	4.44	5.20	6.11	7.20	8.51	17.6%
Total	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 34 LATIN AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.83	0.93	1.03	1.15	1.28	1.43	1.61	1.80	11.8%
Rest of Latin America	0.94	1.04	1.16	1.29	1.44	1.61	1.80	2.02	11.7%
Total	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.3 WHOLE-GENOME SEQUENCING

7.3.1 ACCURATE REFERENCE GENOMES GENERATED FOR MICROBIAL IDENTIFICATION TO BOOST MARKET

Microbial whole-genome sequencing (WGS) has proved to be an important tool for mapping the genomes of novel organisms, finishing genomes of known organisms, or comparing genomes across multiple samples. This technology further helps researchers in generating accurate reference genomes for microbial identification and other comparative genomic studies. For functional studies of the microbiome, WGS metagenomic approaches are more commonly utilized. This is because WGS is affordable and particularly because of an array of computational tools to analyze microbiome sequencing data.

The advantages offered by this technique have resulted in increased adoption among scientists/researchers. As compared to other traditional genomic analysis methods, NGS-based microbial whole-genome sequencing eliminates labor-intensive cloning steps, saves time, and simplifies the workflow. This technique also helps to identify low-frequency variants and genome rearrangements that may be missed or are too expensive to identify using other methods.

TABLE 35 WHOLE-GENOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	26.93	30.39	34.29	38.72	43.75	49.62	56.47	64.54	13.5%
Europe	14.70	16.65	18.84	21.35	24.21	27.55	31.46	36.07	13.9%
Asia Pacific	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%
Latin America	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%
Middle East and Africa	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.08	7.5%
Total	47.29	53.48	60.45	68.39	77.44	87.98	100.31	114.85	13.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 36 NORTH AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	24.40	27.56	31.12	35.17	39.79	45.16	51.43	58.83	13.6%
Canada	2.53	2.83	3.16	3.54	3.97	4.46	5.03	5.70	12.5%
Total	26.93	30.39	34.29	38.72	43.75	49.62	56.47	64.54	13.5%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 37 EUROPE: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.17	3.61	4.11	4.68	5.33	6.10	7.00	8.08	14.5%
UK	2.84	3.24	3.71	4.24	4.85	5.57	6.41	7.42	14.9%
France	2.25	2.55	2.89	3.27	3.72	4.23	4.84	5.56	14.0%
Italy	0.63	0.72	0.81	0.91	1.03	1.17	1.33	1.53	13.5%
Spain	0.49	0.55	0.62	0.70	0.79	0.89	1.01	1.15	13.2%
Rest of Europe	5.32	5.98	6.72	7.55	8.50	9.59	10.86	12.35	12.9%
Total	14.70	16.65	18.84	21.35	24.21	27.55	31.46	36.07	13.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 38 ASIA PACIFIC: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.72	2.00	2.31	2.67	3.09	3.59	4.17	4.88	16.1%
Japan	0.80	0.93	1.09	1.26	1.47	1.71	2.00	2.34	16.6%
India	0.10	0.12	0.14	0.16	0.18	0.21	0.24	0.28	15.6%
Rest of Asia Pacific	1.45	1.67	1.92	2.21	2.54	2.93	3.38	3.92	15.3%
Total	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 39 LATIN AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.44	0.48	0.52	0.57	0.62	0.68	0.75	0.83	9.5%
Rest of Latin America	0.50	0.54	0.59	0.64	0.70	0.76	0.84	0.92	9.2%
Total	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.4 SHOTGUN SEQUENCING

7.4.1 ADVANCEMENTS IN DATA ANALYSIS SOLUTIONS TO SUPPORT ADOPTION

Shotgun sequencing involves breaking up large DNA sequences into multiple small pieces/sequences and then reassembling them into larger contiguous pieces representing the original large DNA sequence. Shotgun sequencing is a relatively new approach used to analyze many organisms present in a given complex sample and comprehensively sequence all genes. This technique provides researchers or end users with the necessary information to evaluate community biodiversity and its related function in the ecosystem.

The advantages offered by shotgun metagenomic sequencing over capillary sequencing, PCR-based approaches, and traditional next-generation sequencing techniques, the growing adoption of shotgun metagenomic sequencing among end users, and the growing number of metagenomic sequencing research activities across the globe are the key factors driving the growth of the shotgun metagenomic sequencing market.

Some of the advantages of shotgun metagenomic sequencing are mentioned below:

- Provides tools to study unculturable microorganisms that are difficult or impossible to analyze
- Eliminates the mapping stages, resulting in faster processing and whole-genome sequencing than clone-by-clone sequencing
- Ability to combine many samples in a single sequencing run and obtain high sequence coverage per sample
- Can detect very low abundance organisms of a microbial community, which may be missed or are too expensive to identify using other traditional methods

The challenges related to data interpretation and analysis in shotgun metagenomic sequencing and the high cost associated with the sequencing of multiple genomes are the key factors restraining the growth of this market. On the other hand, the rapid advancements and developments in informatics software are expected to improve the ease and efficiency of shotgun metagenomic analysis.

TABLE 40 SHOTGUN SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	33.80	38.48	43.79	49.86	56.83	64.99	74.58	85.96	14.4%
Europe	18.31	21.07	24.23	27.90	32.14	37.15	43.09	50.20	15.7%
Asia Pacific	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%
Latin America	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%
Middle East and Africa	0.79	0.88	0.98	1.09	1.21	1.34	1.48	1.65	11.0%
Total	59.09	67.70	77.52	88.83	101.89	117.27	135.44	157.10	15.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 41 NORTH AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	30.64	34.90	39.74	45.29	51.65	59.11	67.88	78.29	14.5%
Canada	3.17	3.58	4.05	4.57	5.18	5.88	6.70	7.67	13.6%
Total	33.80	38.48	43.79	49.86	56.83	64.99	74.58	85.96	14.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 42 EUROPE: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.96	4.57	5.26	6.06	6.99	8.09	9.40	10.96	15.8%
UK	3.54	4.11	4.75	5.51	6.39	7.43	8.67	10.17	16.4%
France	2.80	3.22	3.71	4.27	4.92	5.69	6.60	7.69	15.7%
Italy	0.79	0.91	1.04	1.19	1.37	1.59	1.83	2.13	15.4%
Spain	0.61	0.70	0.80	0.91	1.05	1.21	1.40	1.62	15.3%
Rest of Europe	6.60	7.57	8.67	9.95	11.42	13.14	15.18	17.62	15.2%
Total	18.31	21.07	24.23	27.90	32.14	37.15	43.09	50.20	15.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 43 ASIA PACIFIC: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.13	2.52	2.99	3.53	4.18	4.95	5.89	7.02	18.7%
Japan	0.99	1.18	1.41	1.67	1.99	2.37	2.84	3.41	19.3%
India	0.13	0.15	0.18	0.21	0.25	0.29	0.35	0.41	18.5%
Rest of Asia Pacific	1.79	2.12	2.50	2.95	3.48	4.12	4.89	5.82	18.5%
Total	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 44 LATIN AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.54	0.61	0.68	0.77	0.86	0.97	1.09	1.24	12.7%
Rest of Latin America	0.61	0.68	0.77	0.86	0.97	1.09	1.23	1.39	12.6%
Total	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.5 OTHER TYPES

This segment is inclusive of de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing. Metatranscriptomics is a key technique in metagenomics, which allows researchers to study the function and activity of a complete set of transcripts (RNA sequences) from environmental samples.

The growing number of metagenomic research activities is the major factor driving the growth of this market segment. Over the last few years, the focus on understanding the function and relation of gut microbiome diversity, predicting resistance to specific antibiotics, understanding host-pathogen immune interactions, quantifying gene expression changes, and tracking disease progression has increased significantly.

De novo sequencing generates an initial genomic sequence of a particular organism without a reference genome. De novo sequencing services are applied in the research of animals, plants, and microorganisms, including phylogenetic studies, analysis of species diversity, genetic markers, and other genomic research. Using de novo sequencing to obtain the genomic information of microbes provides a fresh start for exploring the genetic structure and functions, studying the evolutionary origin of microbial populations, as well as developing potential applications across medicine, disease modeling, agriculture, and the environment.

TABLE 45 OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%
Europe	6.81	7.62	8.52	9.53	10.66	11.97	13.48	15.23	12.3%
Asia Pacific	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%
Latin America	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%
Middle East and Africa	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	21.85	24.46	27.36	30.62	34.29	38.52	43.41	49.11	12.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 46 NORTH AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	11.23	12.59	14.11	15.82	17.76	20.00	22.60	25.65	12.7%
Canada	1.16	1.29	1.43	1.59	1.77	1.97	2.20	2.47	11.5%
Total	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis